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Date:	06/04/2023
Instructor:	Samuel Chukwuemeka
Project:	Power Bill: Residential Rates
Company:	Harrisonburg Electric Commission, Harrisonburg, VA https://cdn.harrisonburgelectric.com/wp-content/uploads/2022/06/residential.pdf
Objectives:	**(a.)** I will calculate the electric Bill of the residents of Harrisonburg, VA for the months November through April. Within each range of specific electric usage, manually using an arithmetic method. **(b.)** I will write a piecewise function of the residential rates. **(c.)** I will recalculate the same electric Bill of the residents of Harrisonburg, VA within each range of specific electric usage algebraically using piecewise function method.
Information:	a. Basic Customer Charge Basic Customer Charge $9.50 per Billing month b. Plus kWh Charge 1W1 for Billing months November through April (winter rates):

	First 800 kWh @ $0.08480 per kWh = $67.84 Excess over 800 kWh @ $0.06930 per kWh

Arithmetic Method:

Numbers to be tested by the months:

November	December	January	February	March	April
0 kWh	657 kWh	992 kWh	1412 kWh	1379 kWh	953 kWh

To calculate the usage for each month we will use this method below:

x represents the power usage for the month

- Power usage less than 800 kWh = Basic customer charge + $0.0848(x)$

Excess usage = $\$0.06930 * (x - 800)$

- Power usage above 800 kWh = Basic customer charge + Bill for usage up to 800 kWh + Excess usage

November:

Usage = **0 kWh**

Bill = $9.50 + $0.0848(0)

Bill = $9.50

December:

Usage = **657 kWh**

Bill = \$9.50 + \$0.0848(657)

Bill = \$9.50 + \$55.7136

Bill = \$65.21 (rounded to two decimals, actual \$65.2136)

January:

Usage = **992 kWh** (\$67.84 = First 800 kWh is @ \$0.0848 per kWh)

Bill = \$9.50 + \$67.84 + \$0.06930(992 – 800)

Bill = \$9.50 + \$67.84 + \$0.06930(192)

Bill = \$9.50 + \$67.84 + \$13.3056

Bill = \$90.65 (rounded to two decimals, actual \$90.6456)

February:

Usage = **1412 kWh** (\$67.84 = First 800 kWh is @ \$0.0848 per kWh)

Bill = \$9.50 + \$67.84 + \$0.06930(1412 – 800)

Bill = \$9.50 + \$67.84 + \$0.06930(612)

Bill = \$9.50 + \$67.84 + \$42.4116

Bill = \$119.75 (rounded to two decimals, actual \$119.7516)

March:

Usage = **1379 kWh** ($67.84 = First 800 kWh is @ $0.0848 per kWh)

Bill = $9.50 + $67.84 + $0.06930(1379 – 800)

Bill = $9.50 + $67.84 + $0.06930(579)

Bill = $9.50 + $67.84 + $40.1247

Bill = $117.46 (rounded to two decimals, actual $117.4647)

April:

Usage = **953 kWh** ($67.84 = First 800 kWh is @ $0.0848 per kWh)

Bill = $9.50 + $67.84 + $0.06930(953 – 800)

Bill = $9.50 + $67.84 + $0.06930(153)

Bill = $9.50 + $67.84 + $10.6029

Bill = $87.94 (rounded to two decimals, actual $87.9429)

Piecewise Function:

C = Charge, $x =$ kWh

Basic customer charge = $9.50

First Piece:

Charge per kWh = $0.0848

C(x) = $0.0848($x$) + $9.50

Second Piece:

First 800 kWh will be charge $67.84

Usage over 800 kWh = $\$0.06930 * (x - 800)$

$$C(x) = \$9.50 + \$67.84 + \$0.06930 * (x - 800)$$

$$C(x) = \$77.34 + \$0.06930x - \$55.44$$

$$C(x) = \$0.06930x + \$21.90$$

$$C(x) = \begin{cases} 0.0848x + 9.50, & if\ 0 \leq x \leq 800 \\ 0.06930x + 21.90, & if\ x > 800 \end{cases}$$

Piecewise Function Method:

Numbers to be tested by the months:

November	December	January	February	March	April
0 kWh	657 kWh	992 kWh	1412 kWh	1379 kWh	953 kWh

November = 0 kWh:

$C(x) = 0.0848x + 9.50,\ if\ 0 \leq x \leq 800$

$C(0) = 0.0848(0) + 9.50$

$C(0) = \$9.50$

December = 657 kWh:

$C(x) = 0.0848x + 9.50,\ if\ 0 \leq x \leq 800$

$C(657) = 0.0848(657) + 9.50$

$C(657) = \$65.21$ (rounded to two decimals, actual $65.2136)

January = 992 kWh:

$C(x) = 0.06930x + 21.90, \; if\ x > 800$

C(992) = 0.06930(992) + 21.90

C(992) = $90.65 (rounded to two decimals, actual $90.6456)

February = 1412 kWh:

$C(x) = 0.06930x + 21.90, \; if\ x > 800$

C(1412) = 0.06930(1412) + 21.90

C(1412) = $119.75 (rounded to two decimals, actual $119.7516)

March = 1379 kWh:

$C(x) = 0.06930x + 21.90, \; if\ x > 800$

C(1379) = 0.06930(1379) + 21.90

C(1379) = $117.46 (rounded to two decimals, actual $117.4647)

April = 953 kWh:

$C(x) = 0.06930x + 21.90, \; if\ x > 800$

C(953) = 0.06930(953) + 21.90

C(953) = $87.94 (rounded to two decimals, actual $87.9429)

Citation:

APA style

Harrisonburg Electric Commission Schedule 1S1/1W1 residential service. HARRISONBURG ELECTRIC COMMISSION SCHEDULE 1S1/1W1 RESIDENTIAL SERVICE. (n.d.). https://cdn.harrisonburgelectric.com/wp-content/uploads/2022/06/residential.pdf